

Weekly Assessment for Delta Operations on ESA and CESA-listed Osmerids

Last updated: *January 12, 2026*

Executive Summary

- First flush conditions were met on 12/23/25, and the action was implemented on 12/25/25.
- The First flush action implementation period will end on 1/7/26, then other entrainment management actions will become active.
- Delta smelt are primarily distributed west of the confluence, in Suisun Marsh
- No Delta smelt or longfin smelt salvage has been observed this water year
- Turbidity in the central/south Delta is high

Operational and Regulatory Conditions

- See current Weekly Fish and Water Operations Outlook document.
- Additional information also available on the [SacPAS SMT page](#).

Delta smelt

Biological

- **Delta smelt life stages:** Adult, Juvenile
- **Abundance estimate:** 6821 (95% CL: 1,477 to 20,185) as of the week of December 29, 2025–January 2, 2026
- **Releases:** A total of 163,349 cultured Delta smelt have been released for WY 2026. The most recent release of 24,606 fish occurred in Sacramento River at Rio Vista on Dec 16, 2025.
- **Delta smelt count:** 33 adult Delta smelt and 24 juvenile Delta smelt have been detected this water year. See Table 1 for recent detections, Figure 1 for spatial distribution, and Figure 2 for temporal distribution.
- **Delta smelt salvage:** 0 Delta smelt have been salvaged, and the cumulative seasonal salvage is 0.

Notes

- Since there are few recent detections of Delta smelt, estimation of distribution within the Delta is limited.

- As mentioned in EDSM reporting, fork length ranges reported for Delta smelt and longfin smelt life stages are defined by permit reporting purposes and are not intended to delineate cohorts or distinguish from hatchery or wild origin. See Table 1 caption for fork-length ranges for age groups of Delta smelt.
- See [SacPAS SMT Page](#) for additional details on releases and detection in surveys and salvage.
- Historical salvage trends can be found at: [SacPAS Salvage Timing](#)

Table 1: Delta smelt detections in the last 2 weeks. Fork Length > 58mm = Adult, Fork Length 20-58mm = Juvenile, Fork Length < 20mm = Larva.

Survey	Date	Region	Stratum	Life Stage	Catch
EDSM	2026-01-02	West	Suisun Marsh	Adult	2
EDSM	2026-01-02	West	Suisun Marsh	Juvenile	1
EDSM	2026-01-07	West	Suisun Marsh	Adult	1
EDSM	2026-01-07	West	Suisun Marsh	Juvenile	1
EDSM	2026-01-08	West	Suisun Marsh	Adult	3
EDSM	2026-01-08	West	Suisun Marsh	Juvenile	1

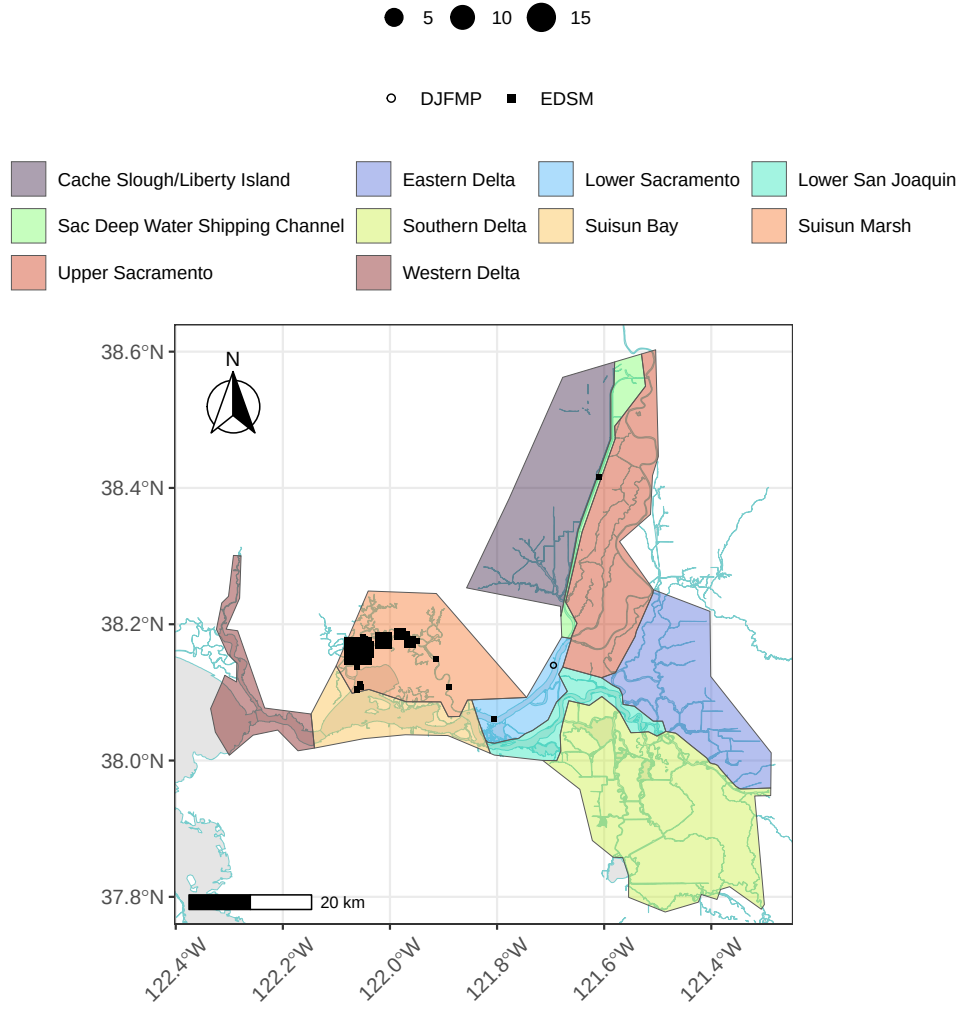


Figure 1: Delta smelt distribution for WY 2026

Table 2: Delta smelt water year totals by life stage

Survey	Region	Life Stage	Total
DJFMP	North	Juvenile	1
EDSM	North	Adult	1
EDSM	West	Adult	32
EDSM	West	Juvenile	23

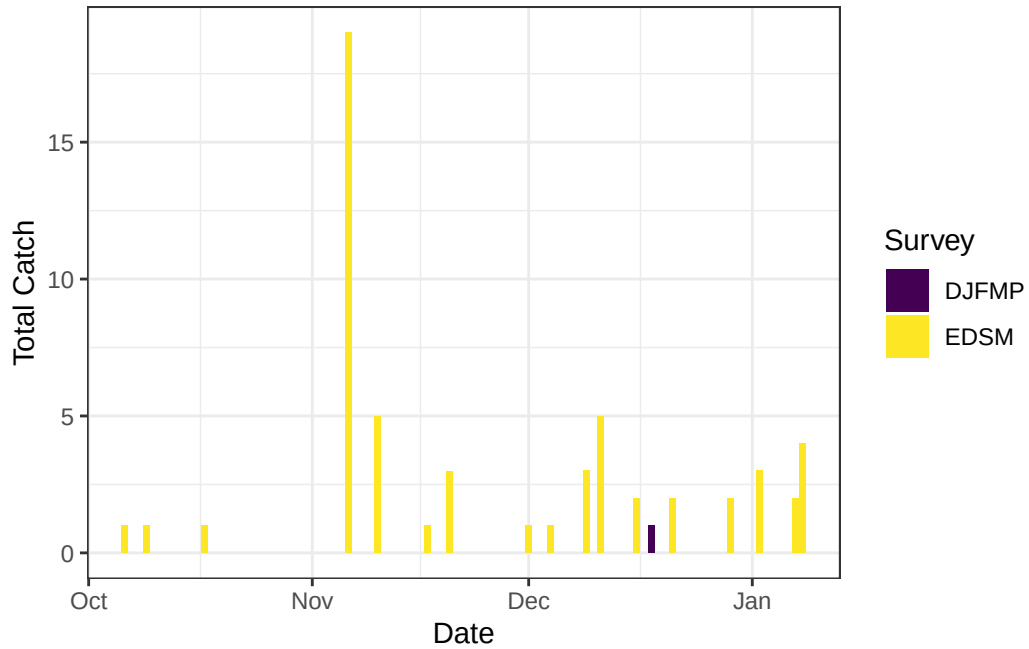


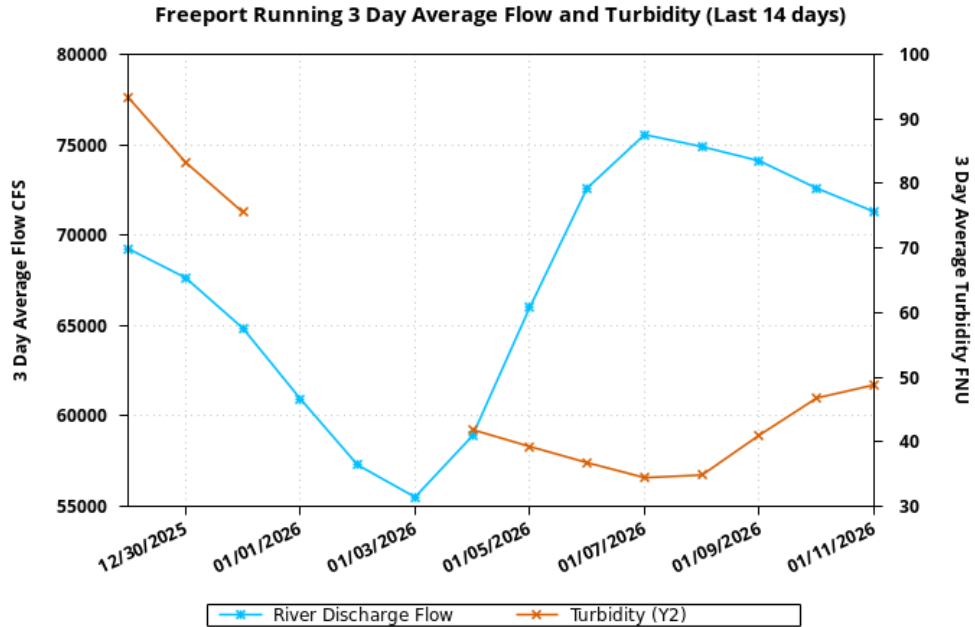
Figure 2: Time series of Delta smelt catch, WY 2026

Environmental

First Flush

Threshold: 3-day avg FPT flow $\geq 25,000$ cfs and 3-day avg FPT turbidity ≥ 50 FNU

- **FPT Flow (3-day average):** 71297 cfs as of Jan 11, 2026
- **FPT Turbidity (3-day average):** 48.81 FNU as of Jan 11, 2026



Preliminary data from CDEC; subject to revision.

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Real-time Assessment Thresholds

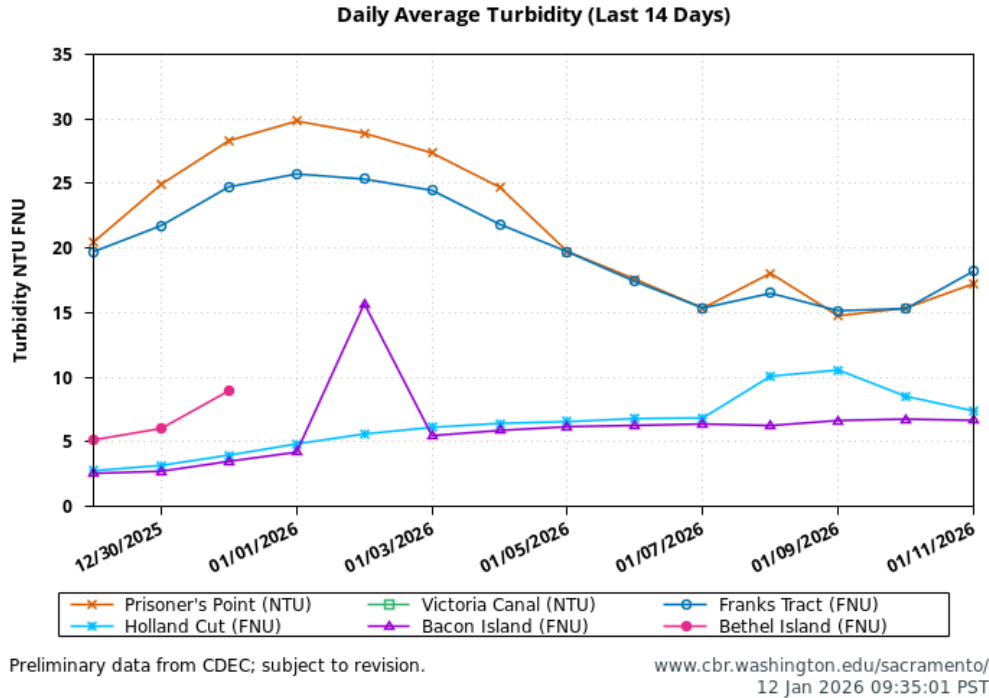
Adult Delta smelt

Threshold: If daily average JPF < 0 AND turbidity $\geq$ 12 FNU at OBI, HOL and OSJ

- **JPF:** 18,657 cfs as of Jan 08, 2026
- **OBI Turbidity:** 6.65, 6.76, 6.66 FNU as of Jan 11, 2026
- **HOL Turbidity:** 10.56, 8.52, 7.39 FNU as of Jan 11, 2026
- **OSJ Turbidity:** 15.15, 15.31, 18.22 FNU as of Jan 11, 2026

Offramp Adult/Onramp Larval and Juvenile Protections when RVB or SJJ > 12°C

- **RVB temperature (3-day average):** 10.19°C as of Jan 11, 2026



- See [Bay-Delta Live](#) for recent Delta-wide turbidity conditions.

Larval/juvenile Delta smelt

Threshold: After the onset of spawning, if JPF < 0 cfs AND turbidity is ≥ 12 FNU in the south Delta AND PTM modeling indicates the action would avoid $\geq 5\%$ entrainment of Delta smelt population after 30 days

- **12-station South Delta Turbidity:** The most recent average turbidity was 19.1 FNU as of Dec 30, 2025

Evaluation

Delta smelt:

1. After the start of entrainment management, is JPF < 0 and is daily average turbidity ≥ 12 FNU in the OMR corridor (stations OBI, HOL, and OSJ)?

No, these conditions will not be met this week.

2. Has the average water temperature at Jersey Point or Rio Vista not exceeded 53.6°F (12°C) for 3 consecutive days and/or has this action already been taken during WY 2026?

Temperature at Rio Vista or Jersey Point has not exceeded the threshold. The Delta smelt adult entrainment management action has not yet been taken in WY 2026.

3. What is the evidence for the onset of Delta smelt spawning?

Upstream migration for Delta smelt occurs between September and December and in response to “first flush” conditions (Sommer et al. 2011, Grimaldo et al. 2009). Migration typically ranges one to four weeks after flow and turbidity increases, based on salvage data (Sommer et al. 2011). Historically, detections of ripe Delta smelt began in January and peaked in February and March and the majority of Delta Smelt spawning occurs within a temperature range of 9-18 °C (Damon et al. 2016). Based on [historical monitoring data](#) from the past few years, first detection of larvae in the Central and South Delta has typically occurred by mid to late March. Based on historic data the migration is likely occurring now, and the onset of spawning has likely not yet occurred.

4. After the onset of spawning, have the following conditions occurred: $\geq 5\%$ entrainment of the Delta smelt population at facilities after 30 days?

The onset of spawning is unlikely to have occurred and JPF is not < 0 cfs.

Longfin smelt

Biological

- **Longfin smelt life stages:** Adult, Juvenile, NA, Larva
- **Longfin smelt count:** 277 adult, 440 juvenile, and 10 larval longfin smelt have been detected this water year. See Table 3 for recent detections, Figure 3 for spatial distribution, and Figure 4 for temporal distribution.
- **Longfin smelt salvage:** 0 longfin smelt have been salvaged, and the cumulative seasonal salvage is 0.

Table 3: Longfin smelt detections in the last 2 weeks. Fork Length $> 84\text{mm}$ = Adult, Fork Length $20\text{-}84\text{mm}$ = Juvenile, Fork Length $< 20\text{mm}$ = Larva.

Survey	Date	Region	Stratum	Life Stage	Catch
DJFMP	2025-12-30	N/A	Chipps Island	Adult	54
DJFMP	2025-12-30	N/A	Chipps Island	Juvenile	1
DJFMP	2025-12-31	N/A	Chipps Island	Adult	44
DJFMP	2025-12-31	N/A	Chipps Island	Juvenile	2
DJFMP	2026-01-05	Bay	Bay	Juvenile	1
DJFMP	2026-01-05	N/A	Chipps Island	Adult	1
DJFMP	2026-01-06	N/A	Chipps Island	Adult	1
DJFMP	2026-01-08	N/A	Chipps Island	Adult	3

Table 3: Longfin smelt detections in the last 2 weeks. Fork Length > 84mm = Adult, Fork Length 20-84mm = Juvenile, Fork Length < 20mm = Larva.

Survey	Date	Region	Stratum	Life Stage	Catch
EDSM	2025-12-30	West	Suisun Bay	Adult	2
EDSM	2025-12-30	West	Suisun Bay	Juvenile	1
EDSM	2025-12-31	Far West	Suisun Bay	Adult	2
EDSM	2025-12-31	Far West	Suisun Bay	Juvenile	1
EDSM	2026-01-05	Far West	Suisun Bay	Adult	2
EDSM	2026-01-05	Far West	Suisun Bay	Juvenile	2
EDSM	2026-01-06	West	Suisun Bay	Juvenile	1
EDSM	2026-01-07	West	Suisun Marsh	Juvenile	1
sls	2025-12-30	Far West	Western Delta	Larva	3
sls	2025-12-30	Far West	Western Delta	NA	4
sls	2025-12-30	West	Lower San Joaquin	Larva	4

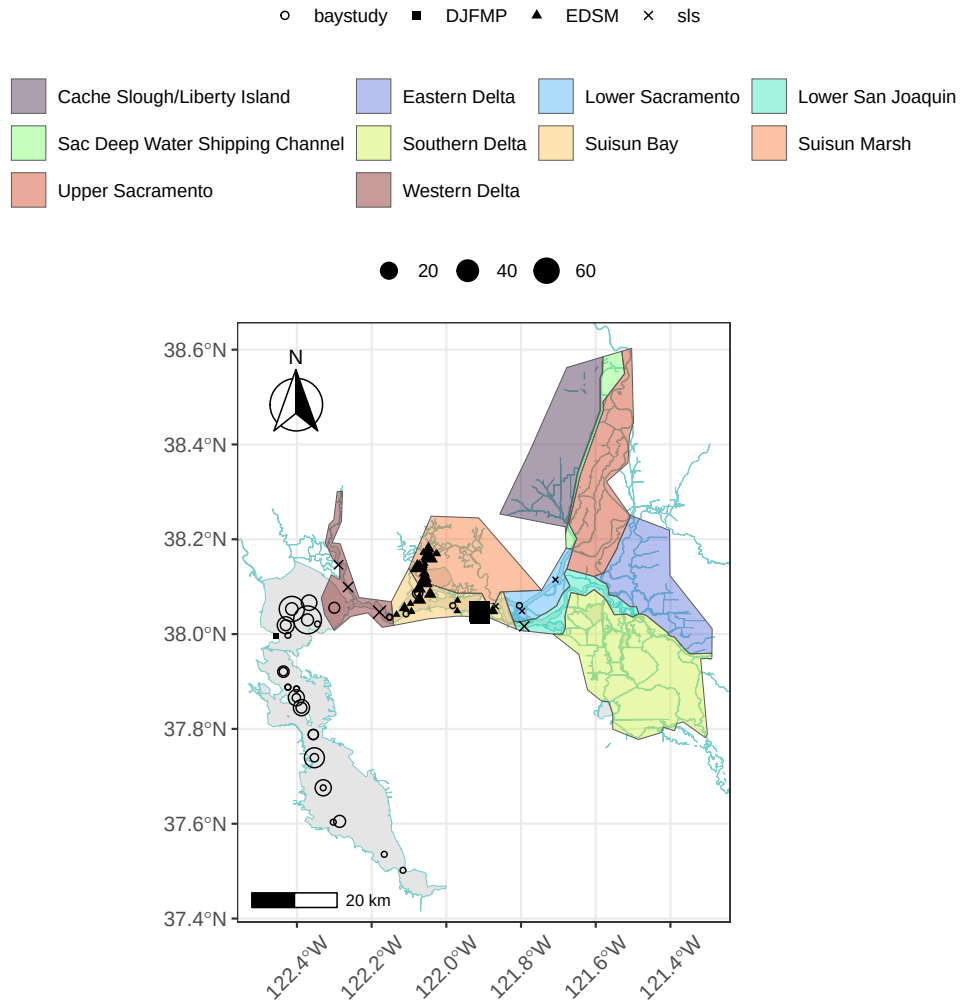


Figure 3: Longfin Smelt Distribution for WY 2026

Table 4: Longfin smelt water year totals by life stage

Survey	Region	Life Stage	Total
DJFMP	Bay	Juvenile	1
DJFMP	N/A	Adult	240
DJFMP	N/A	Juvenile	14
EDSM	Far West	Adult	7
EDSM	Far West	Juvenile	13
EDSM	West	Adult	22
EDSM	West	Juvenile	75

Table 4: Longfin smelt water year totals by life stage

Survey	Region	Life Stage	Total
baystudy	Bay	Adult	6
baystudy	Bay	Juvenile	320
baystudy	Far West	Adult	2
baystudy	Far West	Juvenile	11
baystudy	West	Juvenile	6
sls	Far West	Larva	3
sls	North	Larva	1
sls	West	Larva	6

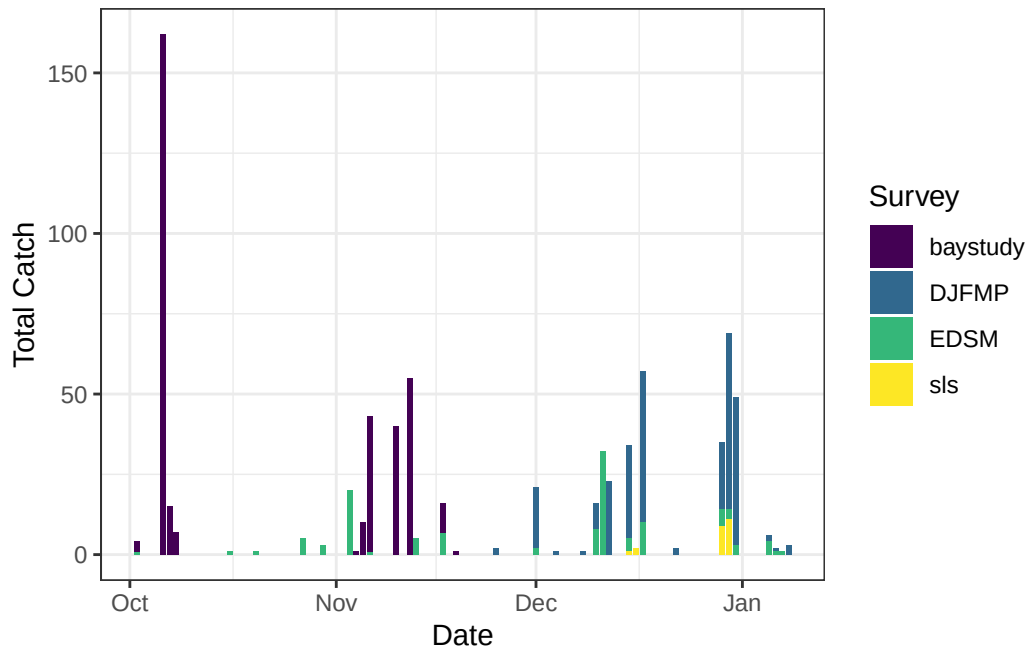


Figure 4: Time series of longfin smelt catch, WY 2026

Real-time Assessment Thresholds

Start of Entrainment Management (Adult Longfin Smelt)

- Not relevant

Adult longfin smelt

- **Threshold:** JPF < 0 cfs, annual loss is on a trajectory to exceed 5% of the adult population abundance, and reduced exports will reduce entrainment in the south Delta
 - Daily average JPF: 18,657 cfs as of Jan 08, 2026
 - Water year total adult longfin smelt salvage = 0

Larval/juvenile longfin smelt

- **Threshold:** JPF < 0 cfs AND population model demonstrates need to reduce entrainment to avoid population decline
 - Daily average JPF: 18,657 cfs as of Jan 08, 2026

Evaluation

Longfin smelt:

1. If JPF < 0, what is the trajectory of annual loss of adult longfin smelt and is it likely to exceed 5% of the adult population estimate? Is South Delta entrainment expected to decrease due to a reduction in export pumping?

JPF is not < 0 cfs and we have not detected any adult longfin smelt in salvage. The ZOI analysis indicates little change in the hydrodynamic footprint between current and forecasted flows this week. The analysis showed a 65 km difference in hydrodynamic footprint between OMRI of -3500 cfs and -5000 cfs, indicating a low risk of entrainment.

2. For larval and juvenile longfin smelt, if JPF < 0 cfs, do particle tracking models show a moderate to high difference in particle fates across different OMRI scenarios? Does Zone of Influence modeling show moderate to high changes in hydrodynamic footprint across different OMRI scenarios? Are these effects anticipated to cause a population decline?

JPF is not less than zero. No PTM was run this week. ZOI modeling shows moderate changes in hydrodynamic footprint between OMRI scenarios.

3. Is there additional information or other analyses that should be considered in this evaluation?

Additional information may be discussed if needed at the DAT call.

End of smelt Entrainment Management

- Not relevant

References

Damon, L. J., S. B. Slater, R. D. Baxter, and R. W. Fujimura. 2016. Fecundity and reproductive potential of wild female Delta smelt in the upper San Francisco Estuary, California. *California Fish and Game* 102(4):188–210.

Grimaldo, L. F., T. Sommer, N. Van Ark, G. Jones, E. Holland, P. B. Moyle, B. Herbold & P. Smith (2009) Factors Affecting Fish Entrainment into Massive Water Diversions in a Tidal Freshwater Estuary: Can Fish Losses be Managed? *North American Journal of Fisheries Management*, 29:5, 1253-1270, DOI: 10.1577/M08-062.1

Sommer, T., F. Mejia, M. Nobriga, and L. Grimaldo. 2011. The Spawning Migration of Delta Smelt in the Upper San Francisco Estuary. *San Francisco Estuary and Watershed Science* 9(2).